EASY · ADVANCED MATH

Advanced Math

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

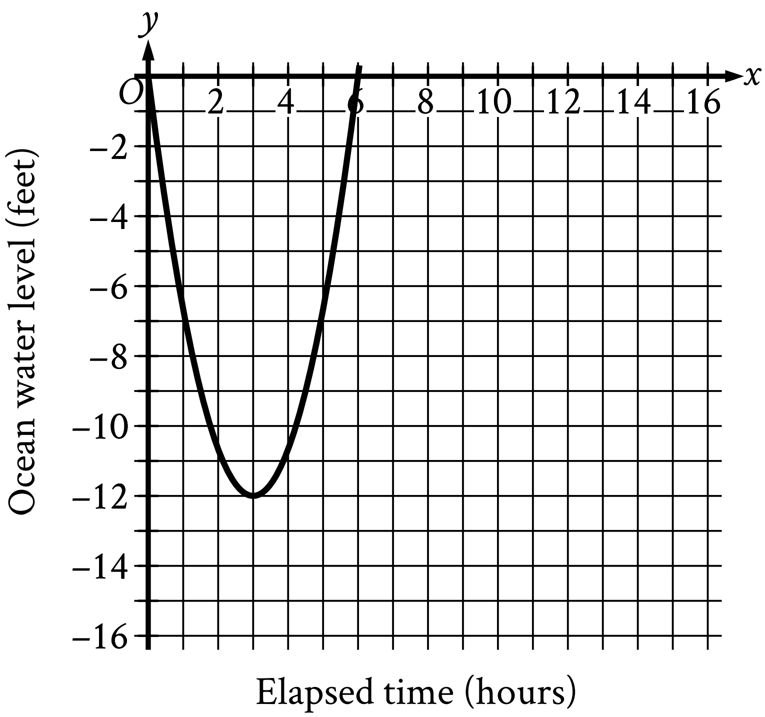
Which expression is equivalent to $5x^5$ - $6x^4 + 8x^3$?

A. $x^4 5x$ - 6

B. $x^3 5x^2$ - $6x + 8$

C. $8x^3 5x^2$ - $6x + 1$

D. $6x^5$-$6x^4 + 8x^3 + 1$



- The parabola opens upward.
- The parabola passes through the following points:
  - (0 comma 0)
  - (3 comma negative 12)
  - (6 comma 0)

Scientists recorded data about the ocean water levels at a certain location over a period of 6 hours. The graph shown models the data, where $y = 0$ represents sea level. Which table gives values of x and their corresponding values of y based on the model?

A.

x	y
0	-12
0	3
3	6

B.

x	y
0	0
3	12
0	-6

C.

x	y
0	0
3	-12
6	0

D.

x	y
0	0
12	3
-6	0

Q3 ADVANCED MATH

$$y - 57 = px$$

The given equation relates the positive numbers p, x, and y. Which equation correctly expresses y in terms of p and x?

A. $y = 57x + p$

B. $y = px + 57$

C. $y = 57px$

D. $y = \frac{px}{57}$

Q4 ADVANCED MATH

Which expression is a factor of $2x^2 + 38x + 10$?

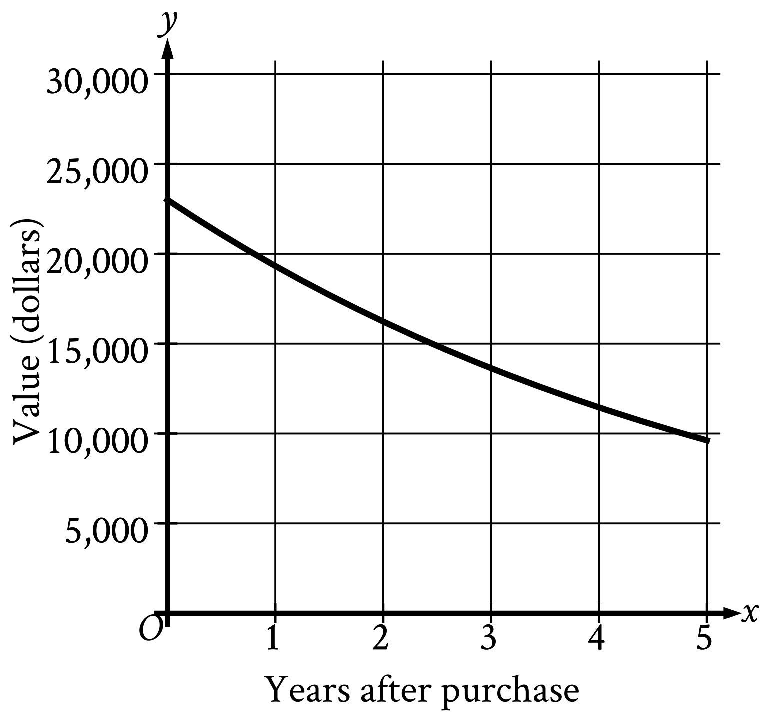
A. 2

B. $5x$

C. $38x$

D. $2x^2$

The graph shows the predicted value y, in dollars, of a certain sport utility vehicle x years after it is first purchased.



- The curve is in quadrant 1.
- The curve trends down gradually from left to right.
- The curve passes through the following approximate points:
    - (0 comma 23,000)
    - (1 comma 19,320)
    - (3 comma 13,632)
    - (5 comma 9,619)

Which of the following is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased?

A. $ 9,619

B. $ 13,632

C. $ 19,320

D. $ 23,000

Q6

ADVANCED MATH

$$y = 5x + 4$$

$$y = 5x^2 + 4$$

Which ordered pair x, y is a solution to the given system of equations?

A. $0, 0$

B. $0, 4$

C. $8, 44$

D. $8, 84$

Q7

ADVANCED MATH

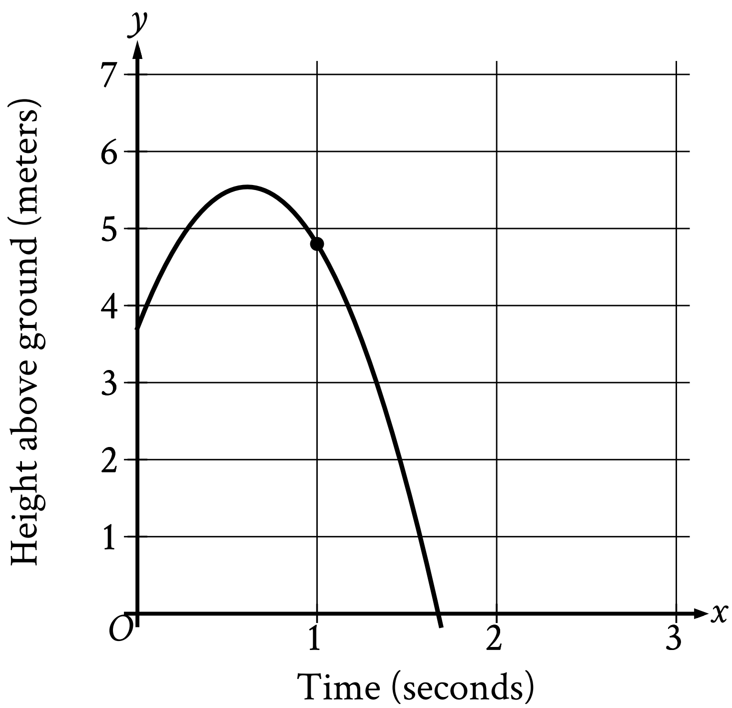
Which expression is equivalent to $34x + 34y$?

A. $34xy$

B. $34(x + y)$

C. $68y$

D. $68x$



- All values are approximate.
- The parabola opens downward.
- The vertex of the parabola is at (0.6 comma 5.5).
- The parabola passes through the following points:
  - (0.0 comma 3.8)
  - (0.6 comma 5.5)
  - (1.0 comma 4.8)
  - (1.7 comma 0.0)

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point $1.0, 4.8$ in this context?

A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.

B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.

C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.

D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.

Q9

ADVANCED MATH

$$x + y = 12$$
$$y = x^2$$

If (x, y) is a solution to the system of equations above, which of the following is a possible value of x?

A. 0

B. 1

C. 2

D. 3

Q10

ADVANCED MATH

Which expression is equivalent to $x^2 + 3x - 40$?

A. $x - 4x + 10$

B. $x - 5x + 8$

C. $x - 8x + 5$

D. $x - 10x + 4$